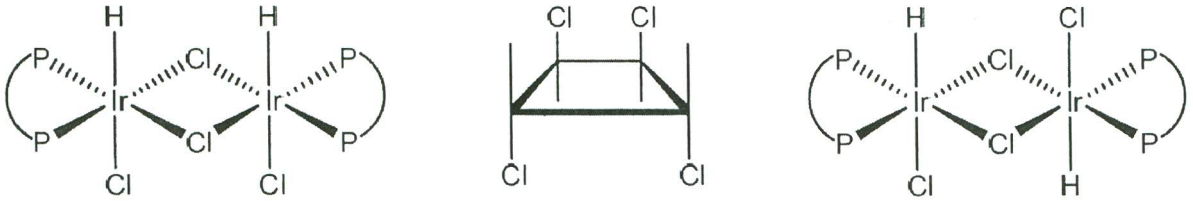
Inorganic Cumulative Exam

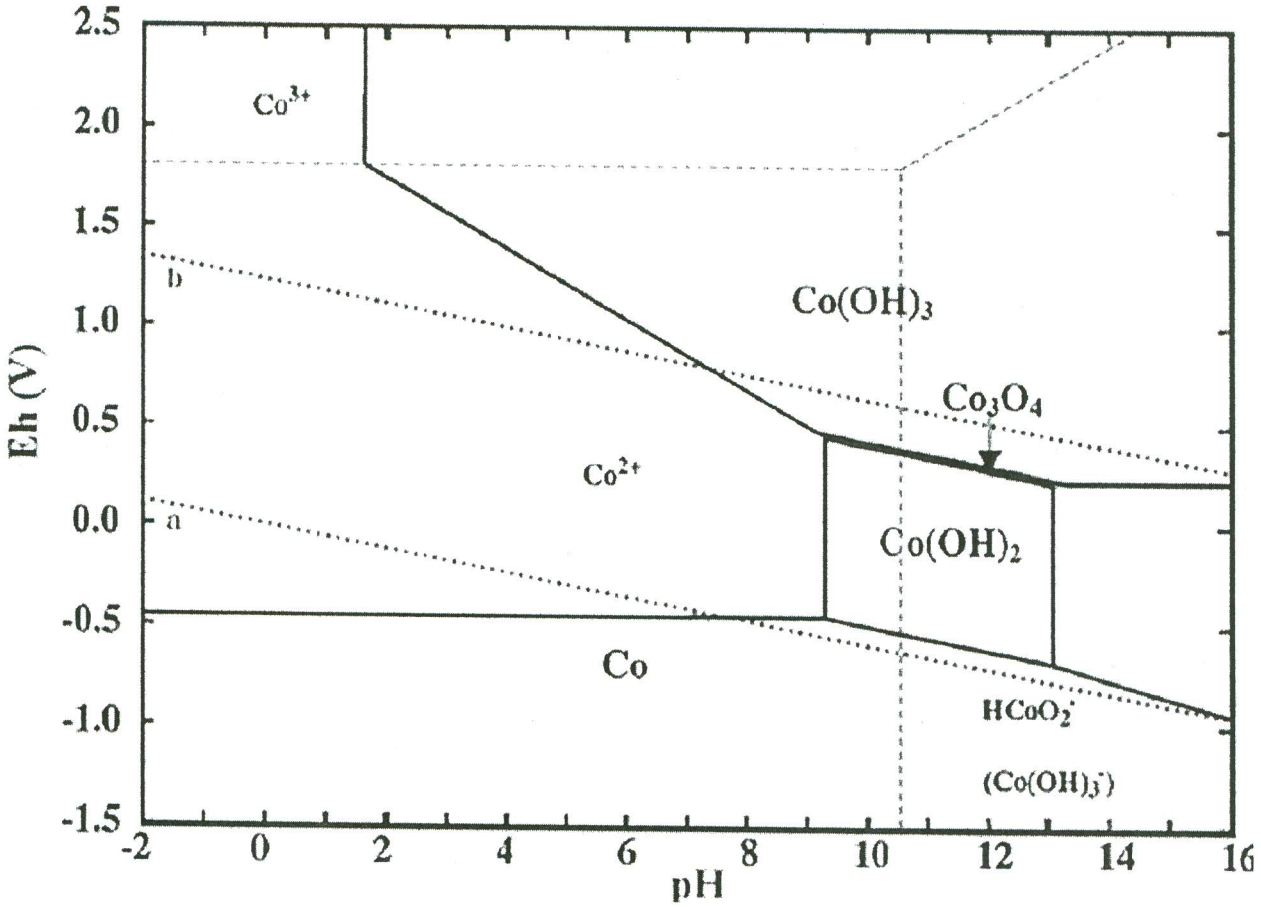
Oct 28 2016

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded.

1. What are the structural differences between amorphous, nematic, and smectic phases in a liquid crystal system?
2. Three parts: (a) (8 pts) Describe a galvanic replacement reaction in the context of nanocrystal synthesis. (b) (8 pts) The shape of some noble metal nanocrystals (Au, Ag) can be controlled by illuminating them during growth. Describe the effect responsible for this. (c) (4 pts) Name another way that the shape of a nanocrystal can be controlled during growth.
3. Describe the concept of a bulk heterojunction in an organic photovoltaic and identify what problem it solves for that application.
4. Of the following six molecules, which have an S_n axis of symmetry? Please draw the S_n axis for each molecule that has one.



Methane, Acetylene, Ferrocene, (and the three in the diagram)

5. Describe the operating principle behind the Peltier effect in thermoelectric materials (ie., electrical cooling). Use band-bending diagrams where necessary.
6. Aluminum (83.8 MJ/L) is more energy dense than coal (~40 MJ/L). Describe a renewable cycle for generating fuels using aluminum.
7. Two parts: (a) (8 pts) What is the hydrophobic effect? Describe a real example from chemistry/biology/etc. of the hydrophobic effect. (b) (12 pts) Describe a spinodal decomposition, and a system+process that induces a spinodal decomposition.

8. Assume you are preparing a magnetic Fe-Co alloy by electrodeposition. Describe three experiments you would perform (and what the data would tell you) to *verify* that the film is a true alloy and not phase segregated.
9. Using crystal field theory and d-orbital diagrams, explain why TiO_2 is white and $TiCl_3$ is violet.
10. Using the following Pourbaix diagram, propose an electrodepositon recipe on a Pt foil that will produce (a) (10 pts) a porous Co metal film, and (b) (10 pts) a porous $Co(OH)_3$ film. A different recipe is likely required for each. Include detailed rationale of parameters like solution concentration, electrode potentials, etc. What effects are you utilizing to generate porosity?